

```
scuts_path = 'e:\SHARE\FE\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
GHFFfitfile = fullfile(scuts_path,'e800GPffCuts.mat');
if ~exist('GHFFfitData','var')
    ld = load(GHFFfitfile);
    GHFFfitData = ld.fit_src_struct;
end
```

```
mi = maps_instrument(787,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);

cut_names = fieldnames(GHFFfitData);
for i=1:numel(cut_names)
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_instrument(mi);
    GHFFfitData.(cut_names{i}) = GHFFfitData.(cut_names{i}).set_sample(sample);
end
```

```
dir_name = "GP";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning

if ~exist('fit_reEi800GP','var')
    res_name = 'FitEn_Ei800ff_1D_dirGP_dE10_constCutBg.mat';
    if isfile(res_name)
        ld = load(res_name);
        fit_reEi800GP = ld.fit_reEi800GP;
    else
        fit_reEi800GP = struct();
    end
end

for i=1:numel(cut_names)
    acolor k
    fprintf('*****\n')
    fprintf('**** CUT: %s\n',cut_names{i});
    fprintf('*****\n')

    cut_to_work = GHFFfitData.(cut_names{i});
    if isfield(fit_reEi800GP,cut_names{i})
        continue;
    end
    plot(cut_to_work); keep_figure;
    all_en = cut_to_work.data.p{2};
    en_range = min_max(all_en);
    cut_en = 100+half_dE:10:en_range(2)-half_dE;
```

```

[fr_800const_bg,res_name] = fit_multicuts_along_direction(...
    {cut_to_work},'FitEn_Ei800ff_1D',dir_name, ...
    1,cut_en,dE_step,half_dE,true);
fit_reEi800GP.(cut_names{i}) = fr_800const_bg;

acolor g
bg_data = extract_bg_par(fr_800const_bg.all_fit_par,[1,2]);
% original slope parameters

pd(bg_data(1));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure; ly 0 0.5;

pd(bg_data(2));lx(fr_800const_bg.all_fit_par(1).en,fr_800const_bg.all_fit_par(end
).en);keep_figure

save(res_name,'fit_reEi800GP');
end

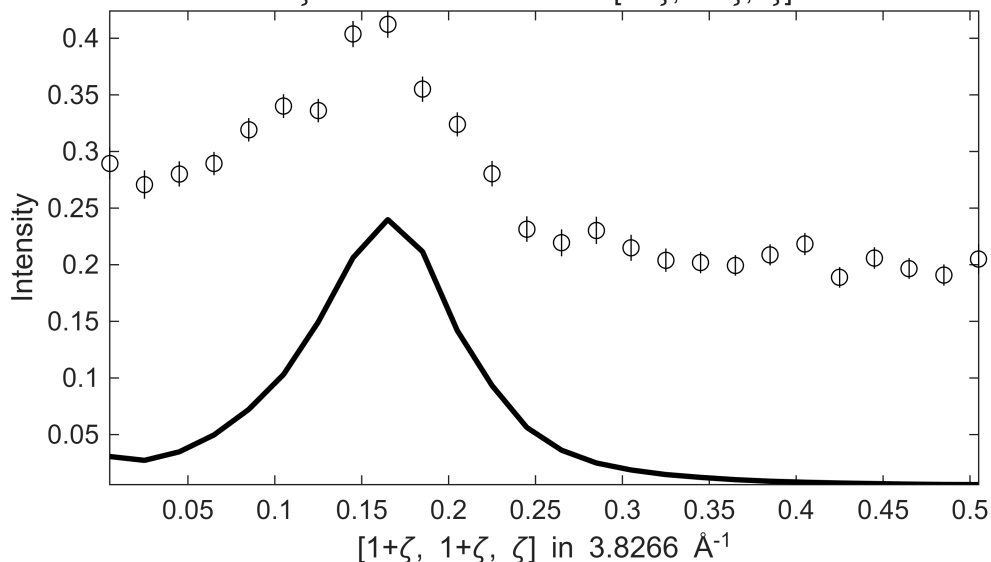
```

```

*****
**** CUT: Ei800_off110minFFGP
*****
**** CUT: Ei800_off200maxFFGP
*****
**** CUT: Ei800_off200minFFGP
*****
**** CUT: Ei800_off110maxFFGP
*****
**** En = 105±5 going to 295
*****

```

Fe_ei800, w2_800of110maxFF.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $100 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

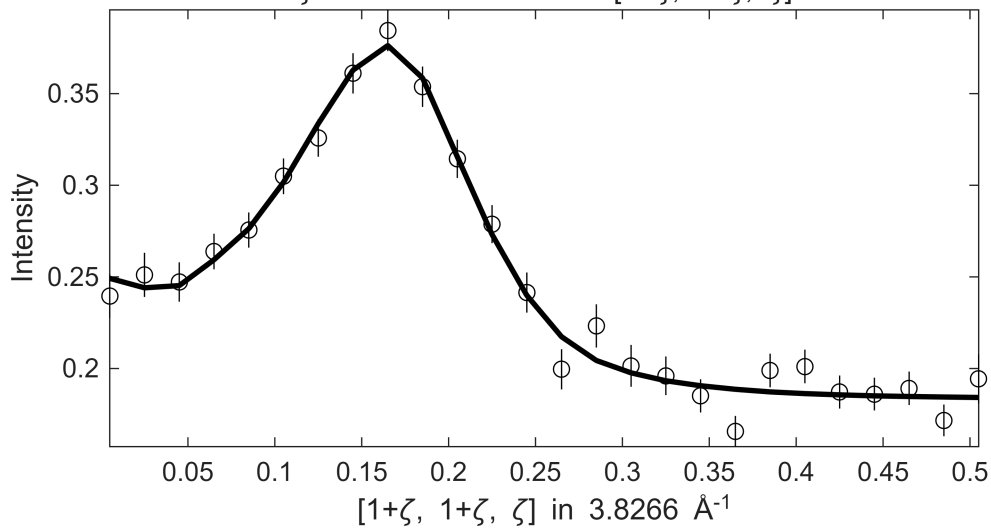


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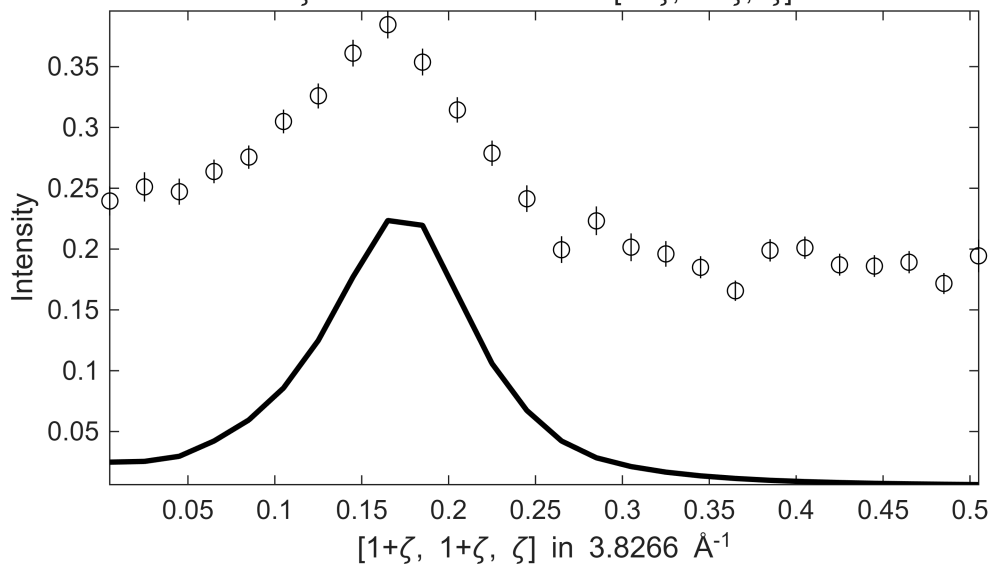
*****
**** En = 115±5 going to 295
*****

```

Fe_ei800, w2_800of110maxFF.sqw
 $S=0.94\pm0.026$; $J0= 36\pm0.55$; $\gamma=49.9372\pm0.914674$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $110 \leq E$
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $110 \leq E$
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

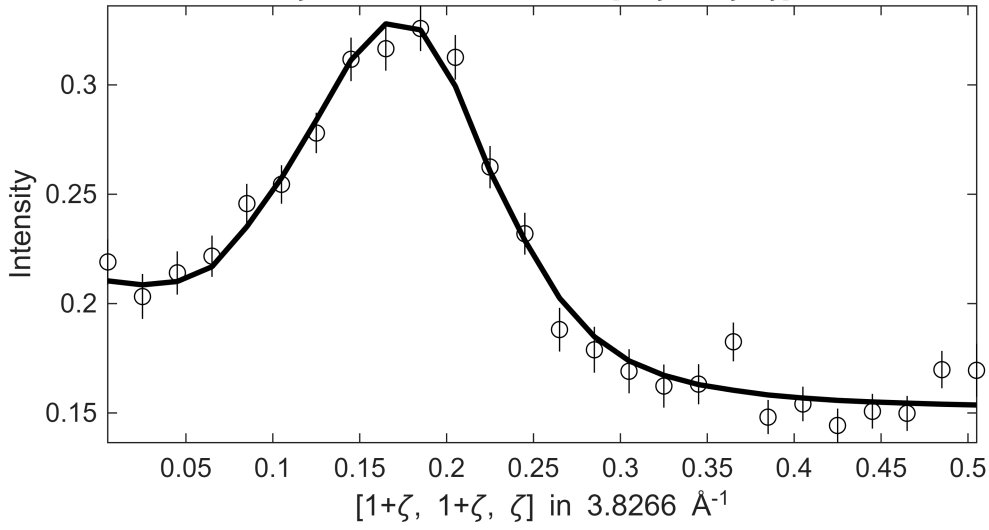


 **** En = 125±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

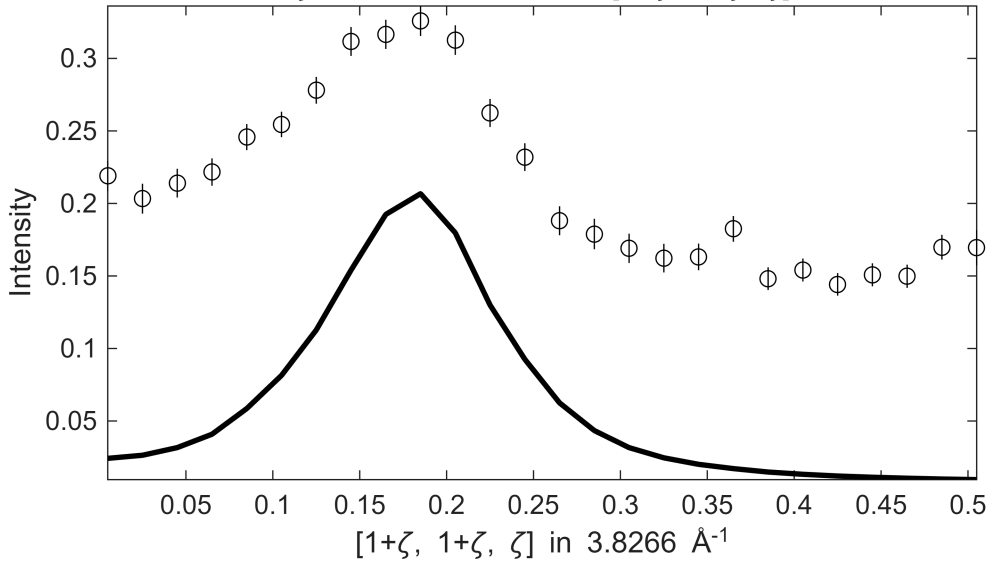
S= 1.1±0.034; J0= 38±0.57; gamma=66.7293±1.13279

$|\xi| \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $120 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

$|\xi| \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $120 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

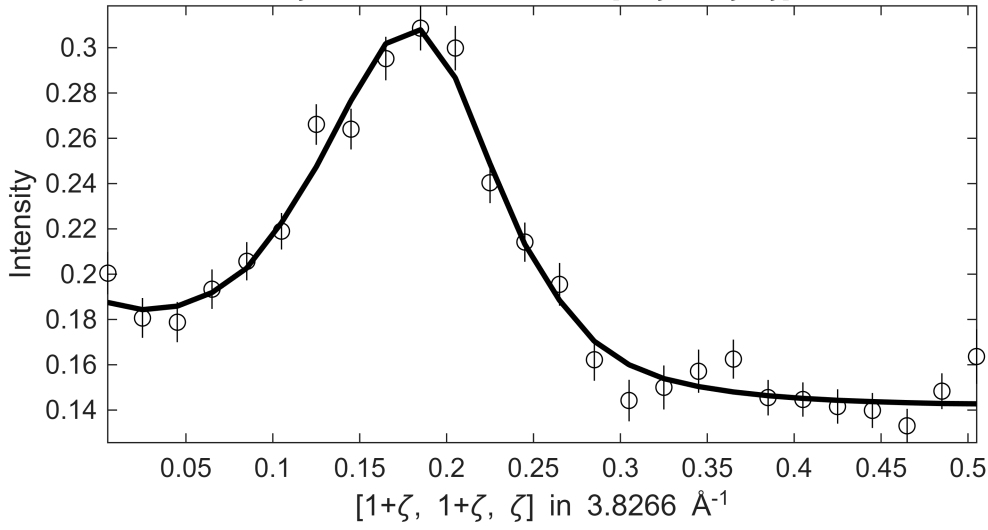


*** En = 135±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

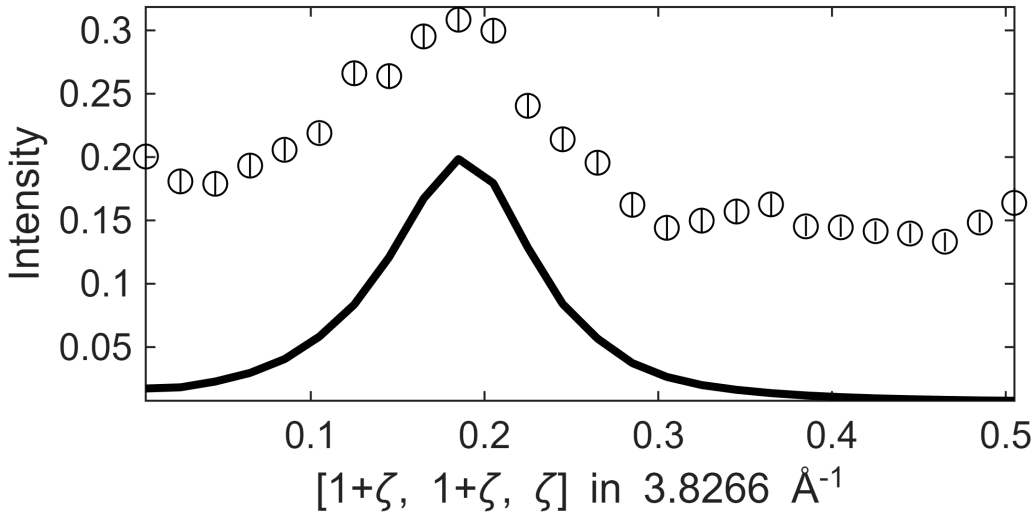
S=0.91±0.094; J0= 37±0.78; gamma=53.6215±7.76878

$|\xi| \leq 0.1$ in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$, $130 \leq E$
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

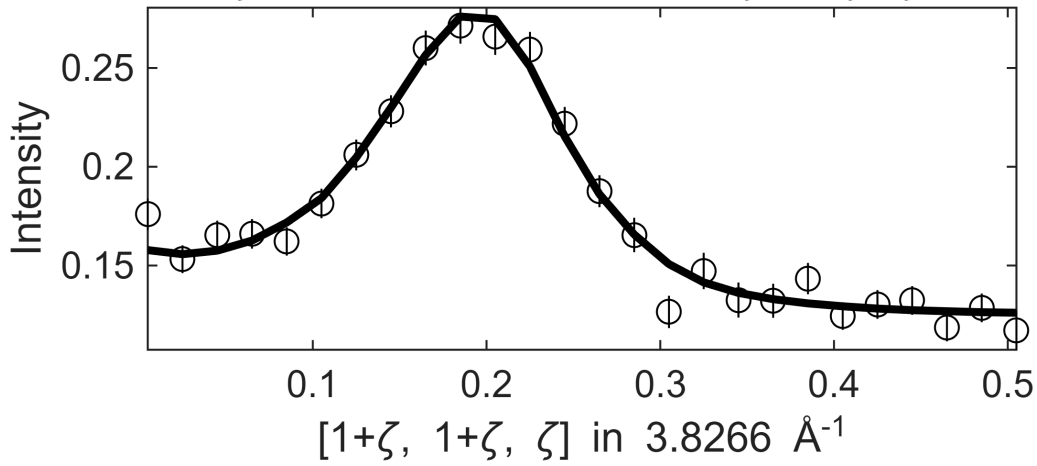


*** En = 145±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

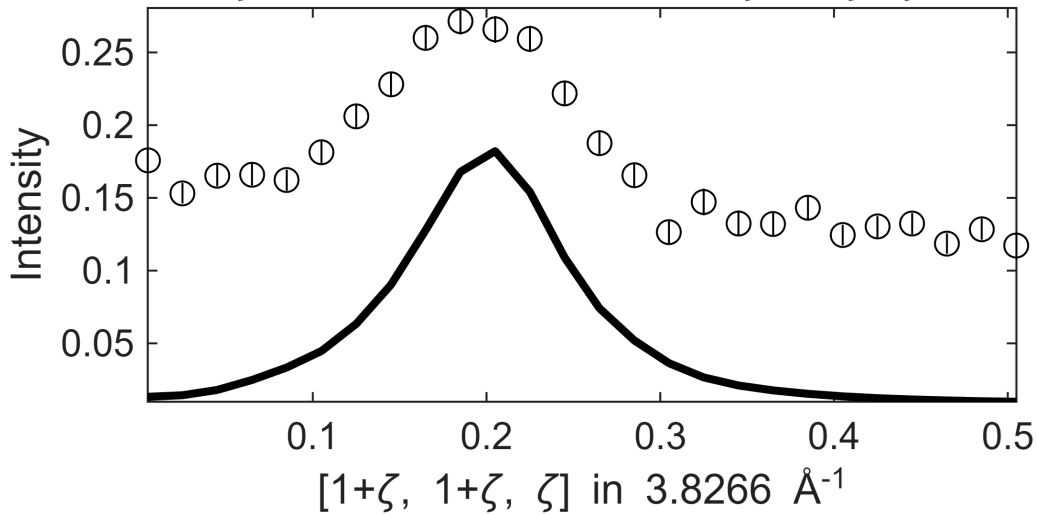
S=0.89±0.029; J0= 36±0.47; gamma=55.716±1.05454

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

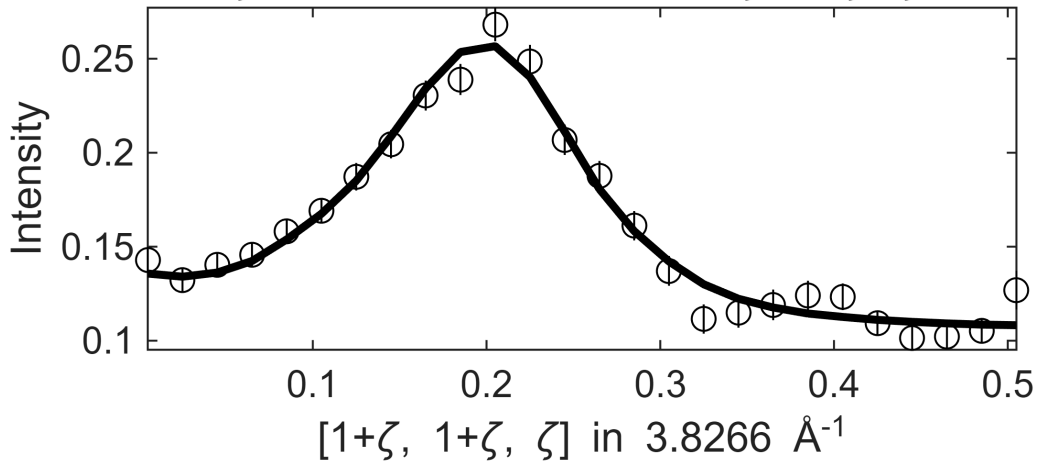


**** En = 155±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

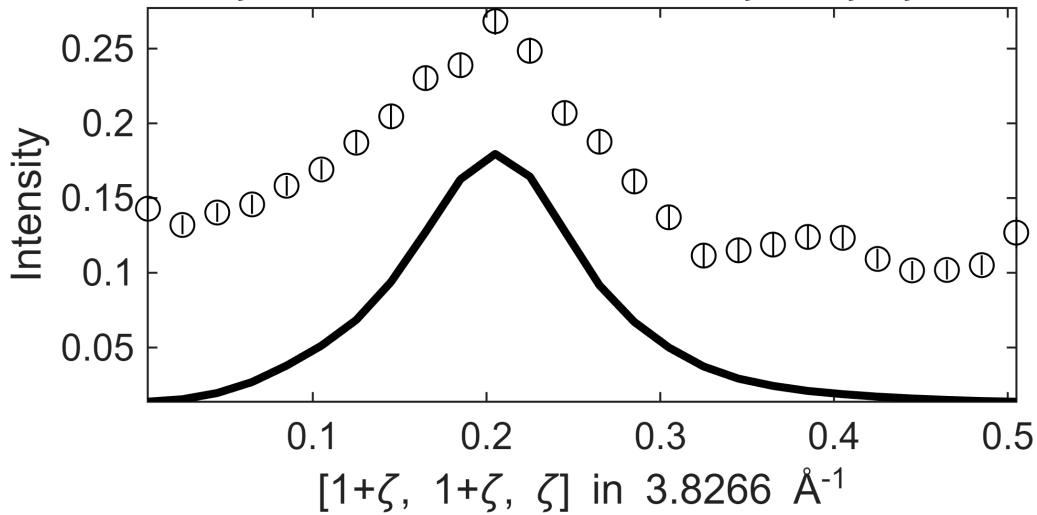
S= 1.1 ± 0.036 ; J0= 38 ± 0.46 ; gamma= 68.7433 ± 1.17219

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

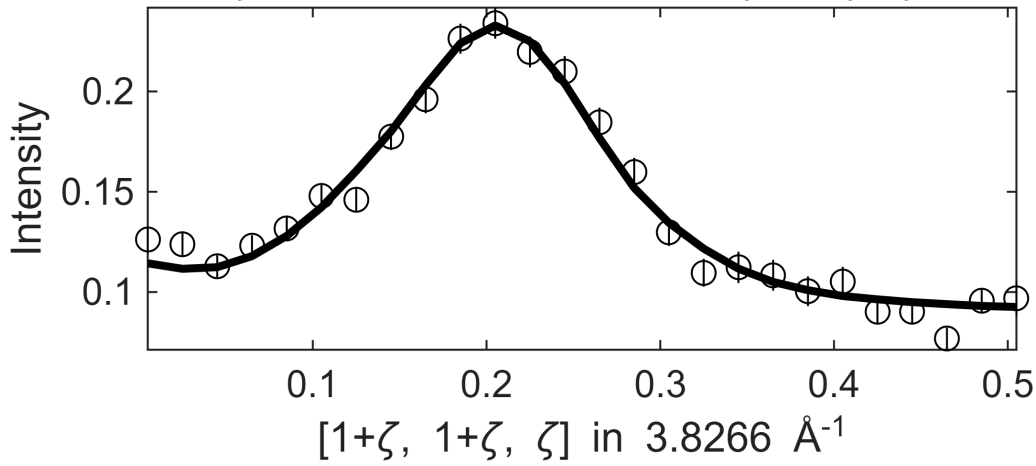


**** En = 165 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

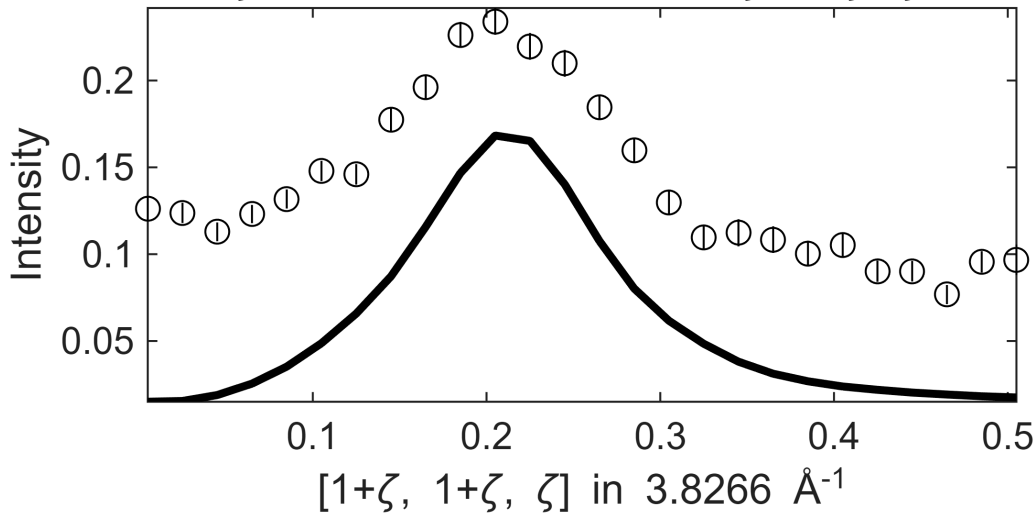
S= 1.2 ± 0.05 ; J0= 39 ± 0.49 ; gamma= 80.3727 ± 1.5033

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



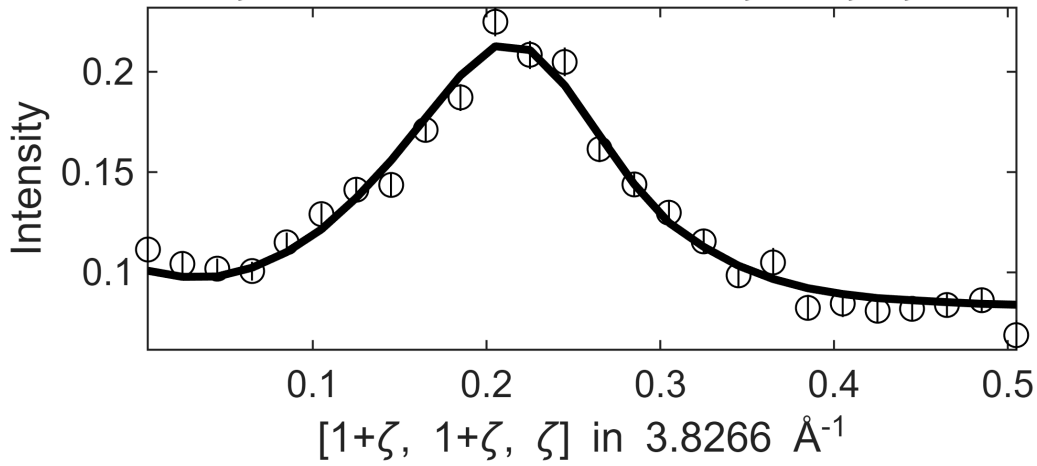
**** En = 175 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

S= 1.1 ± 0.14 ; J0= 39 ± 0.8 ; gamma= 76.154 ± 11.4196

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

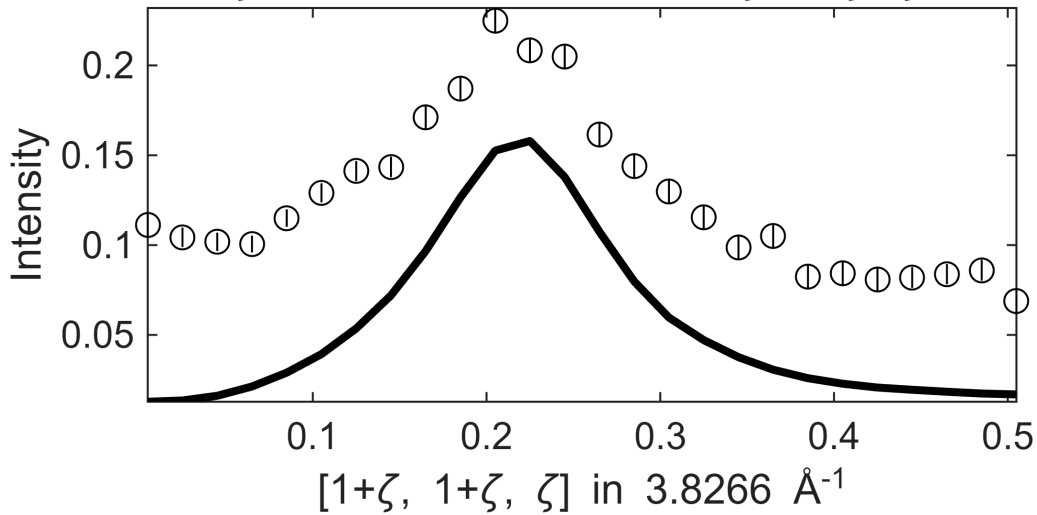
$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



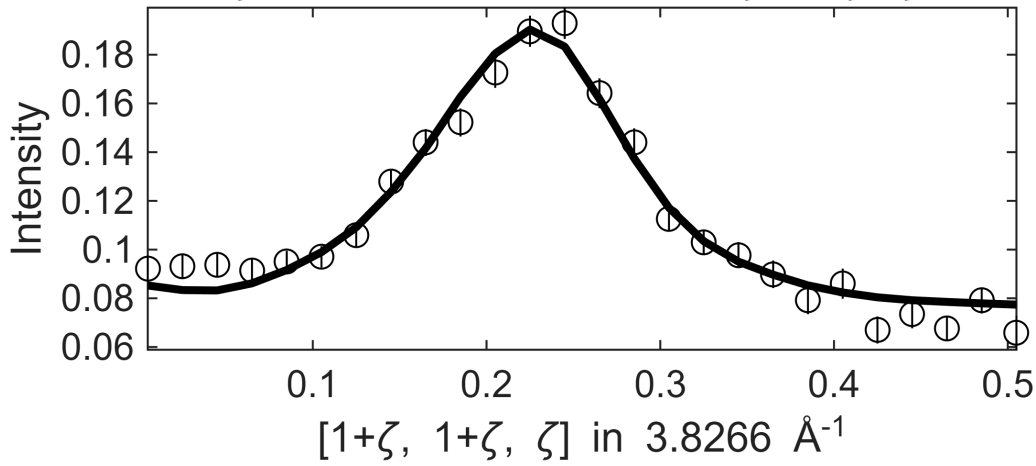
**** En = 185 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

$S=0.83\pm0.036$; $J_0=39\pm0.46$; $\gamma=62.3803\pm1.30135$

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

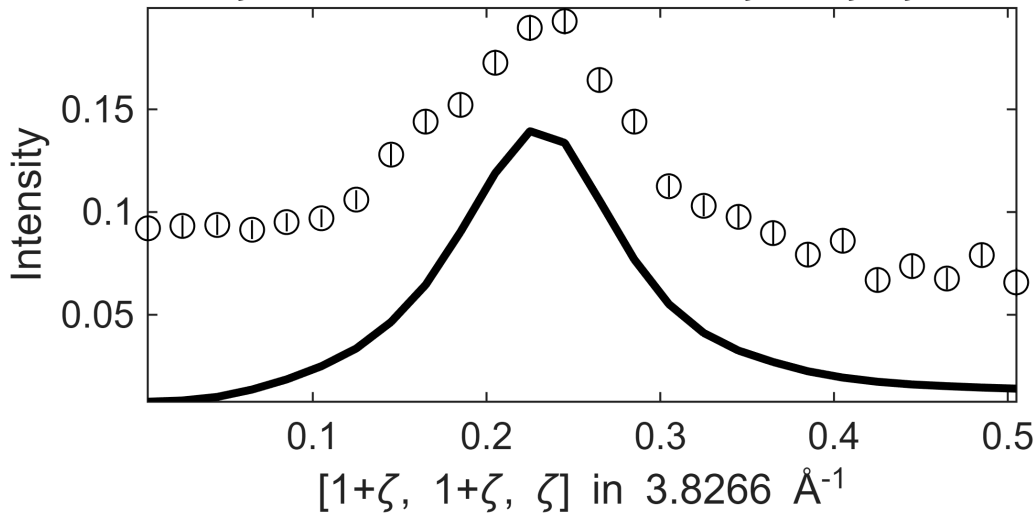
$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

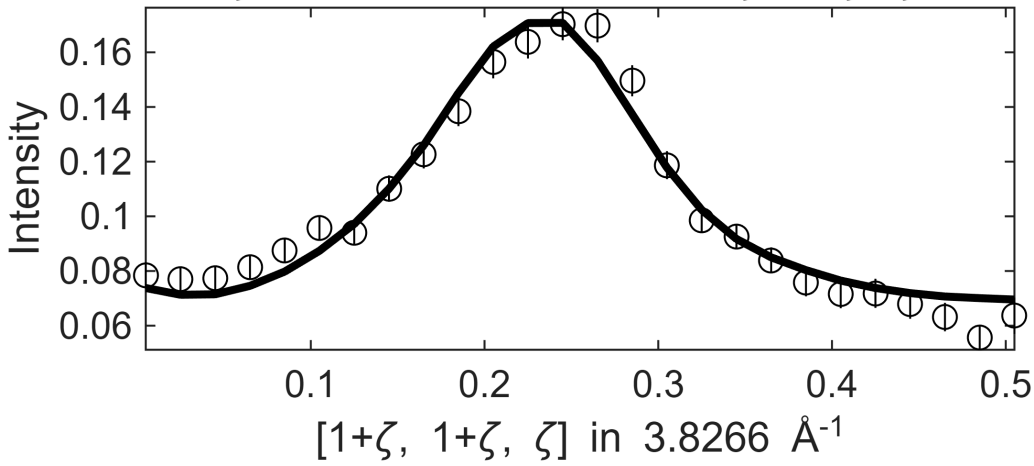


**** En = 195 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

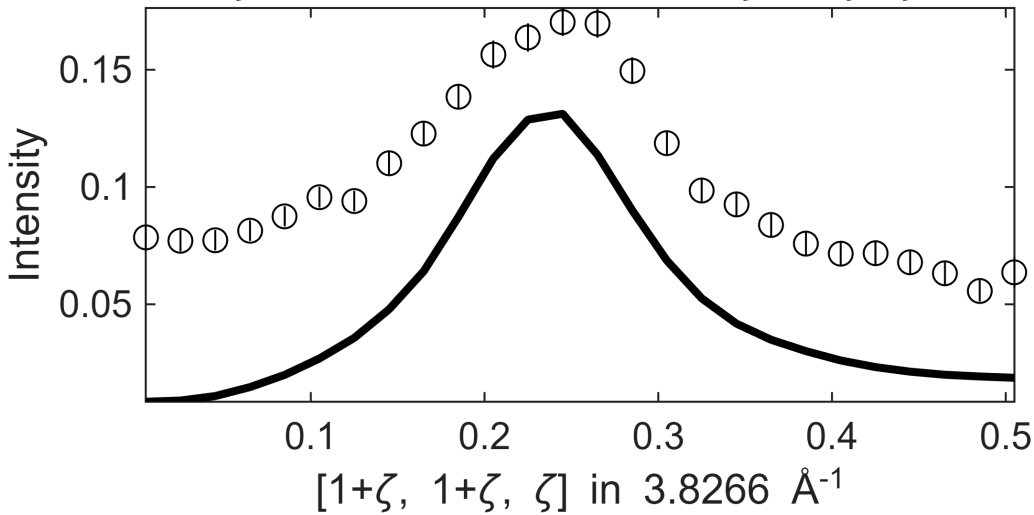
S=0.94±0.14; J0= 39±0.74; gamma=74.0113±11.6616

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

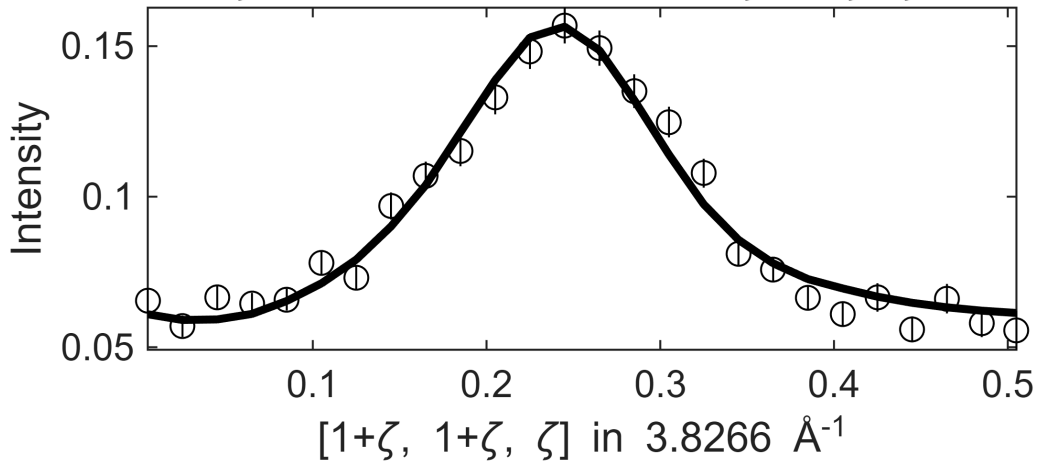


**** En = 205±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

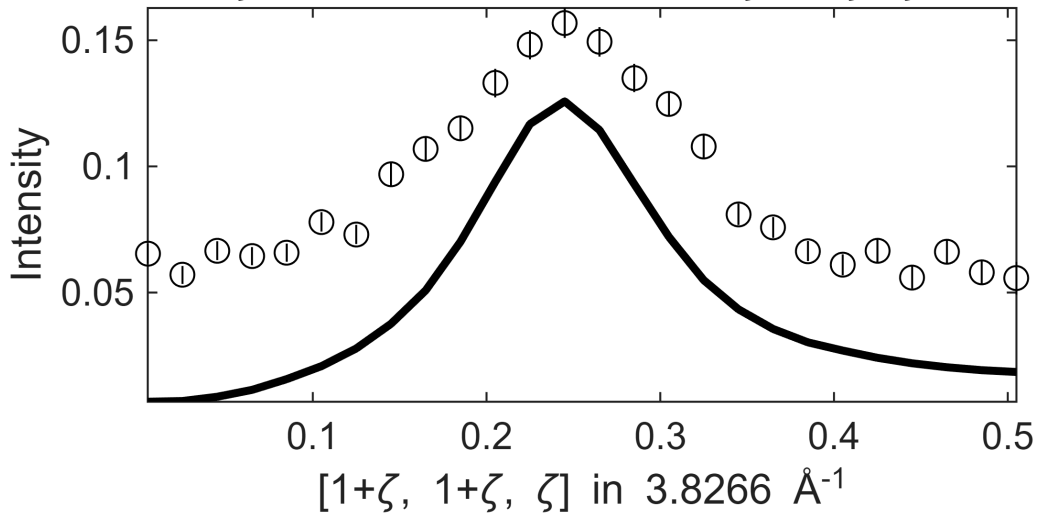
S=0.86±0.043; J0= 39±0.39; gamma=69.6889±1.83291

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

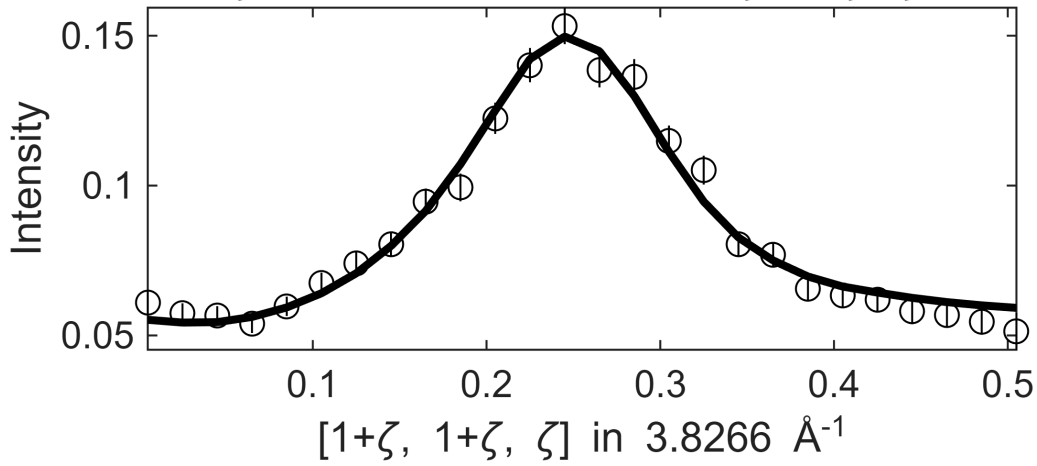


**** En = 215±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

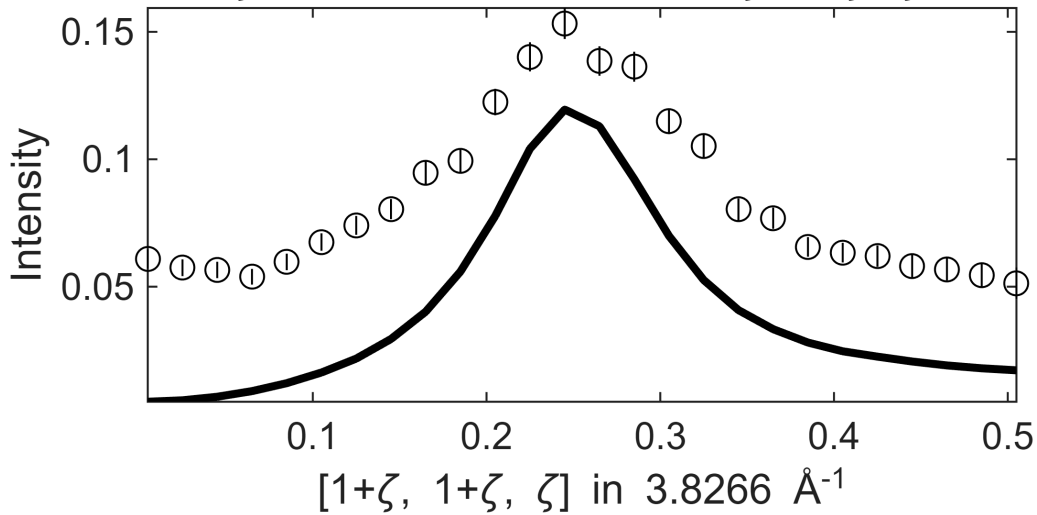
S=0.76±0.069; J0= 40±0.42; gamma=63.4412±6.72751

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

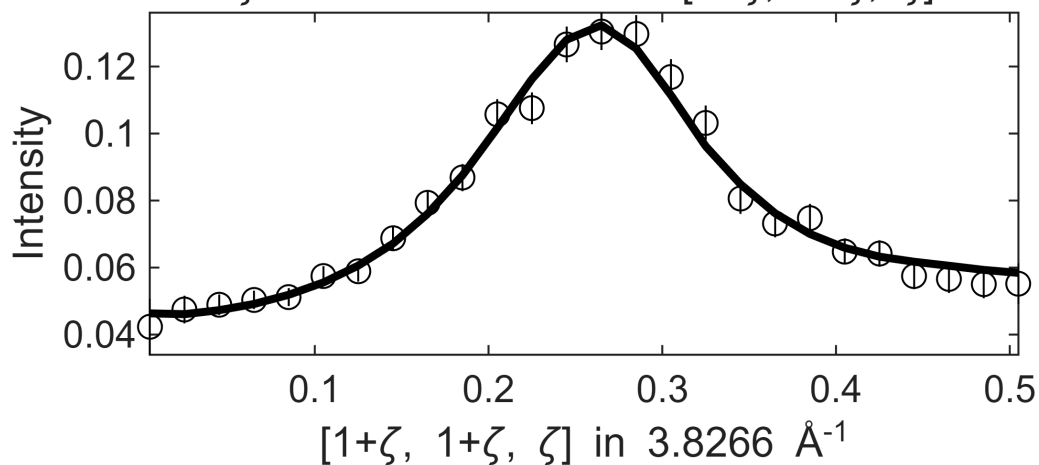
in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



**** En = 225±5 going to 295

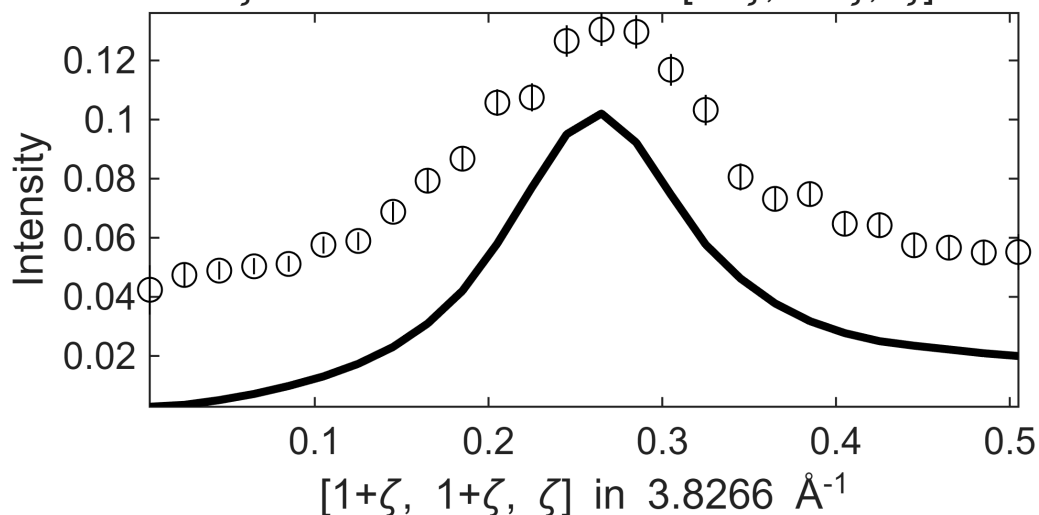
Fe_ei800, w2_800of110maxFF.sqw

S= 0.7 ± 0.02 ; J0= 40 ± 0.25 ; gamma= 66.6249 ± 1.03864
in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



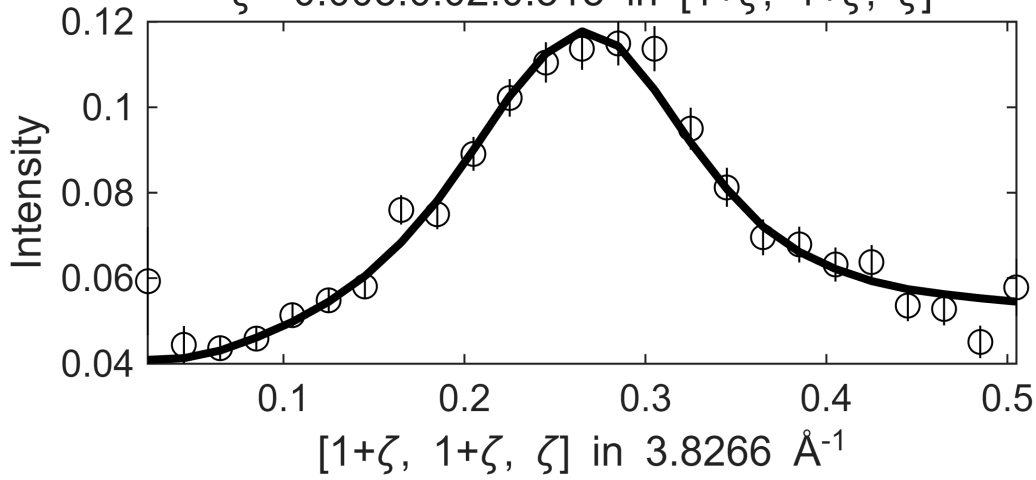
**** En = 235 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

S=0.68±0.03; J0= 40±0.35; gamma=73.4285±1.55616

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

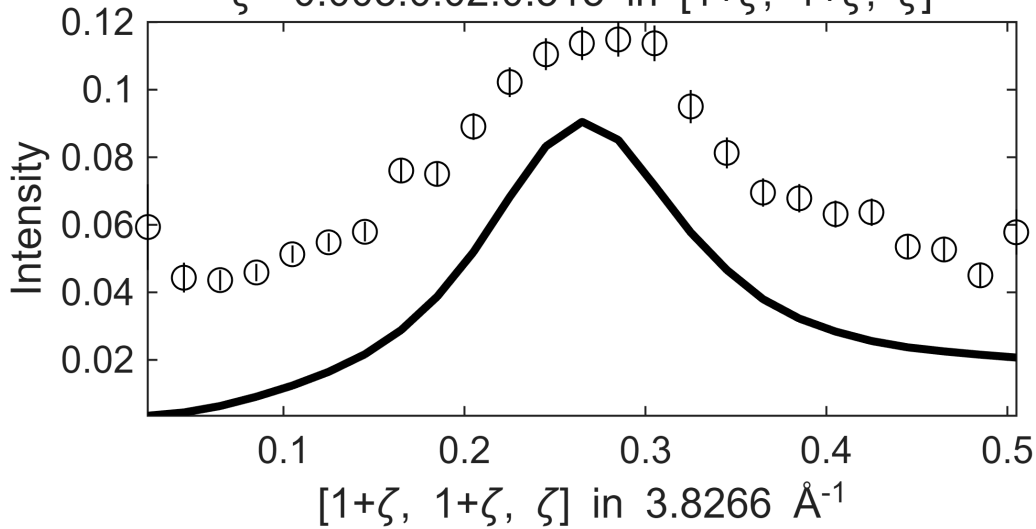
$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

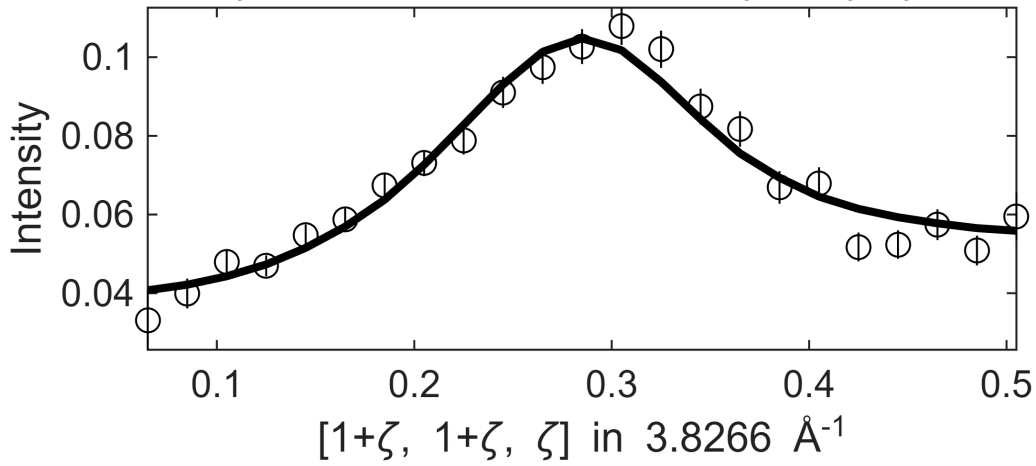


**** En = 245±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

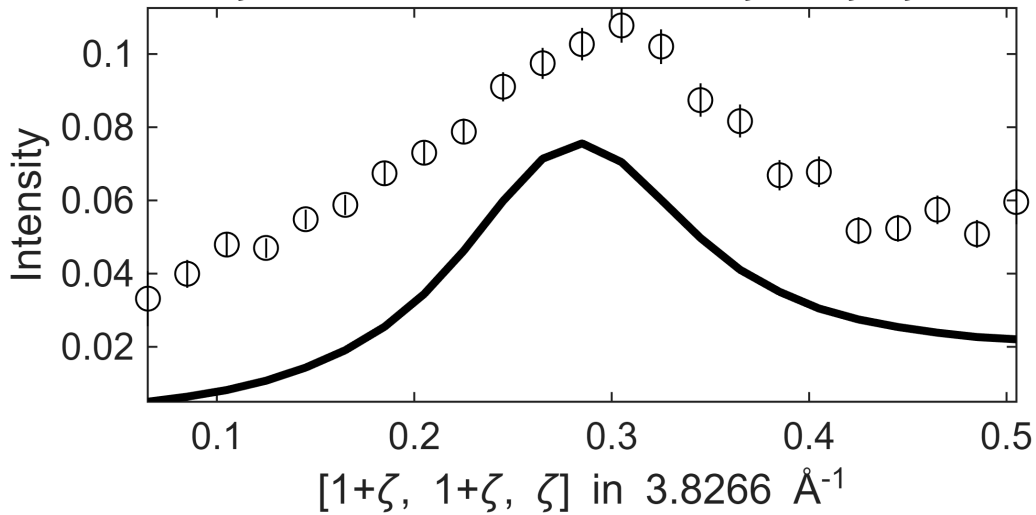
S=0.56±0.032; J0= 40±0.39; gamma=68.5592±1.59013

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



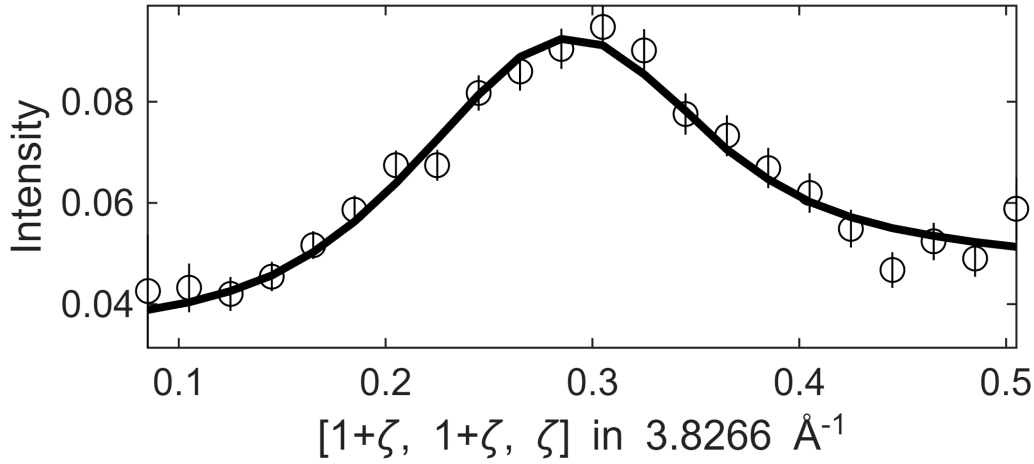
**** En = 255±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

S=0.54±0.032; J0= 40±0.34; gamma=73.9577±1.6366

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

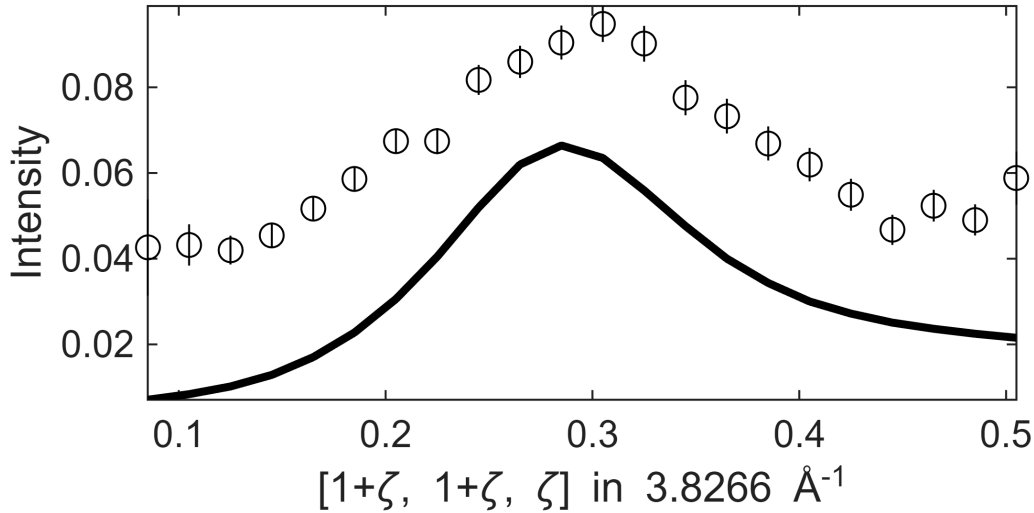
$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

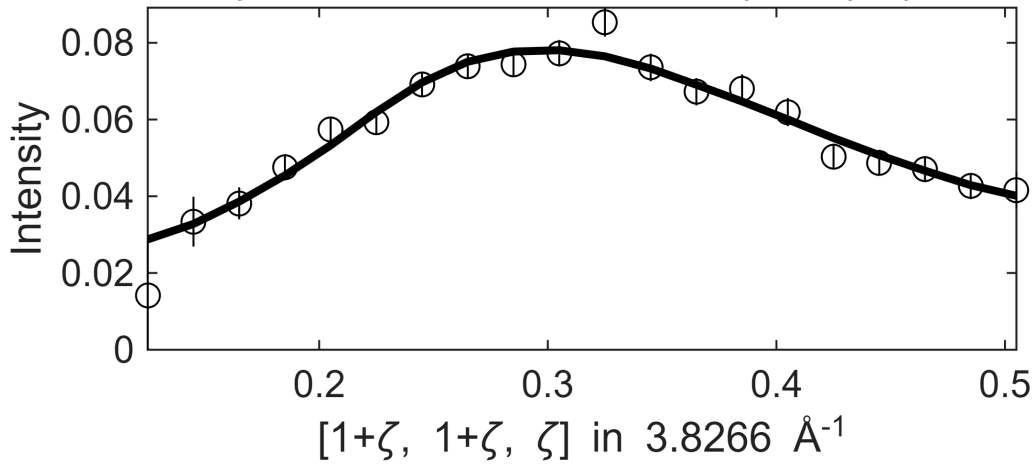


**** En = 265±5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

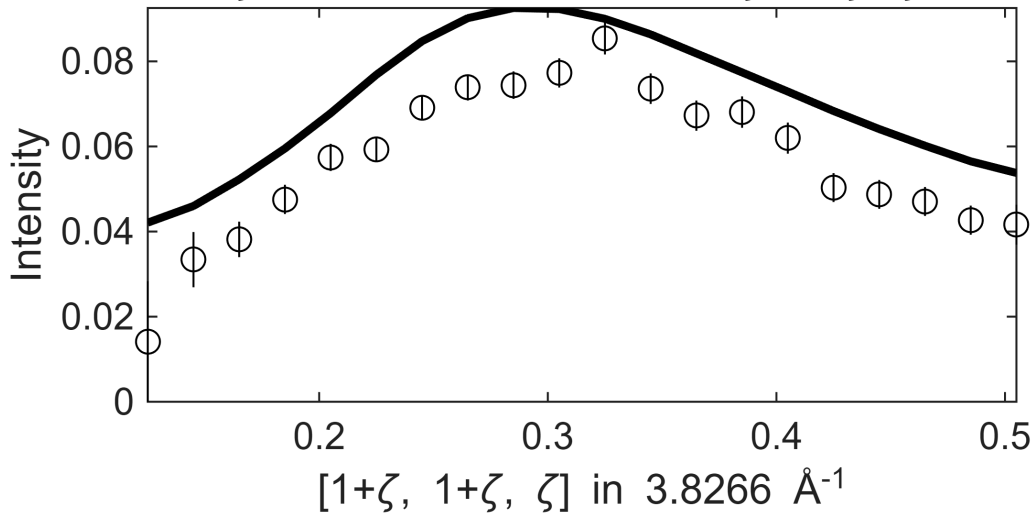
S= 2.2 ± 0.1 ; J0= 43 ± 0.88 ; gamma= 224.759 ± 20.2178

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



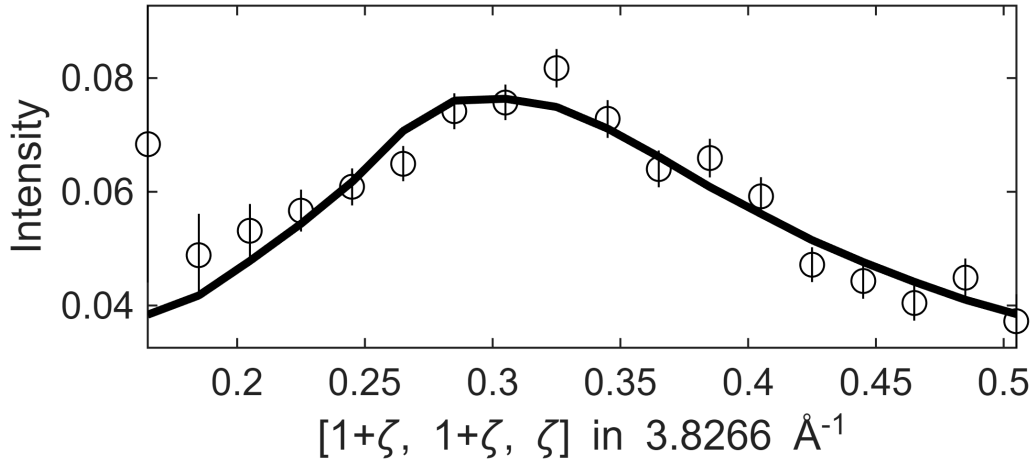
**** En = 275 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

S= 1.3 ± 0.12 ; J0= 42 ± 0.92 ; gamma= 144.23 ± 19.5643

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

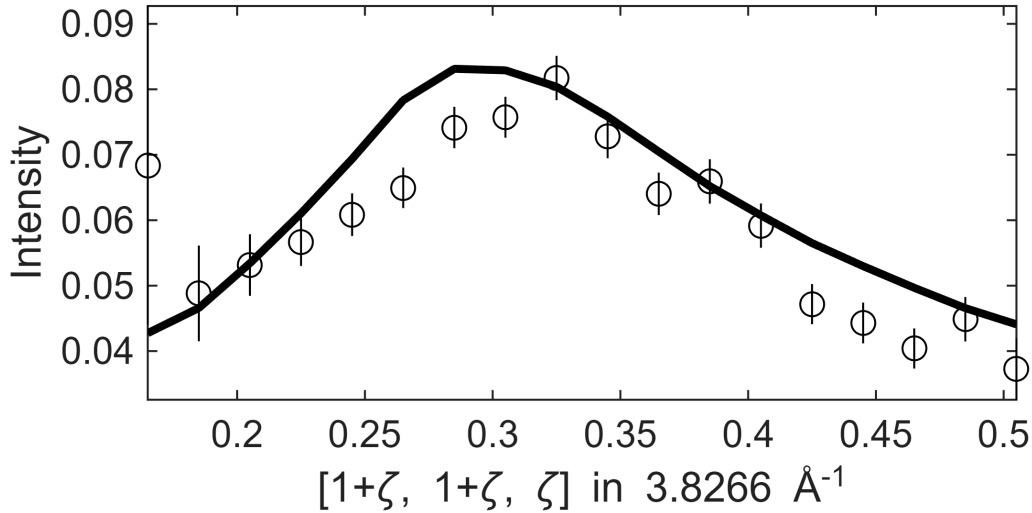
$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

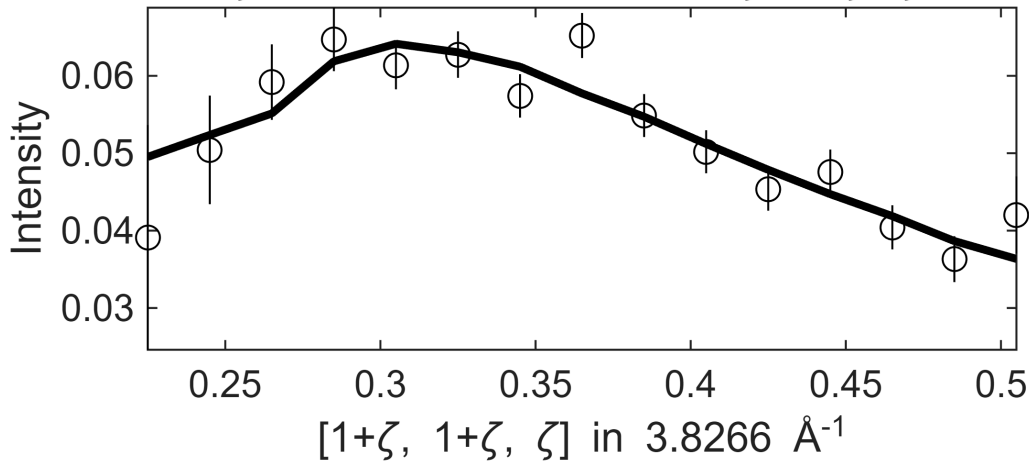


**** En = 285 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

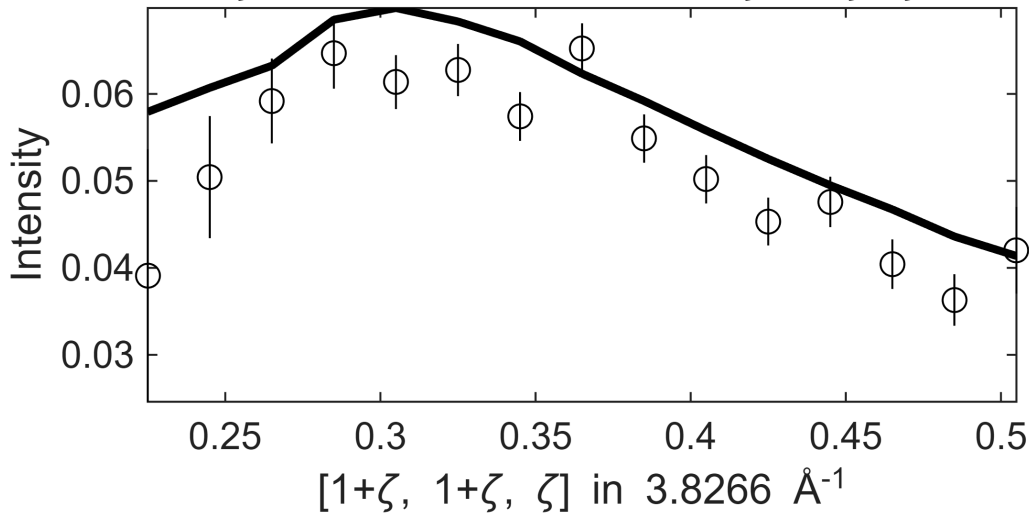
S= 1.8 ± 0.54 ; J0= 44 ± 5.2 ; gamma= 241.852 ± 113.795

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,
 $\zeta=-0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



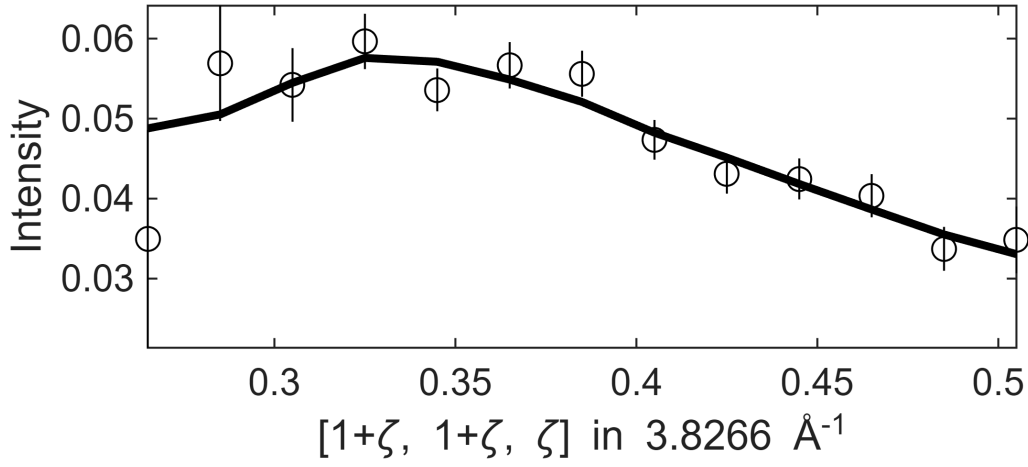
**** En = 295 ± 5 going to 295

Fe_ei800, w2_800of110maxFF.sqw

S= 1.7 ± 0.32 ; J0= 42 ± 4.6 ; gamma= 207.765 ± 77.6047

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$



Fe_ei800, w2_800of110maxFF.sqw

in $[1-\xi, 1+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1-0.5\eta, 1-0.5\eta, \eta]$,

$\zeta = -0.005:0.02:0.515$ in $[1+\zeta, 1+\zeta, \zeta]$

